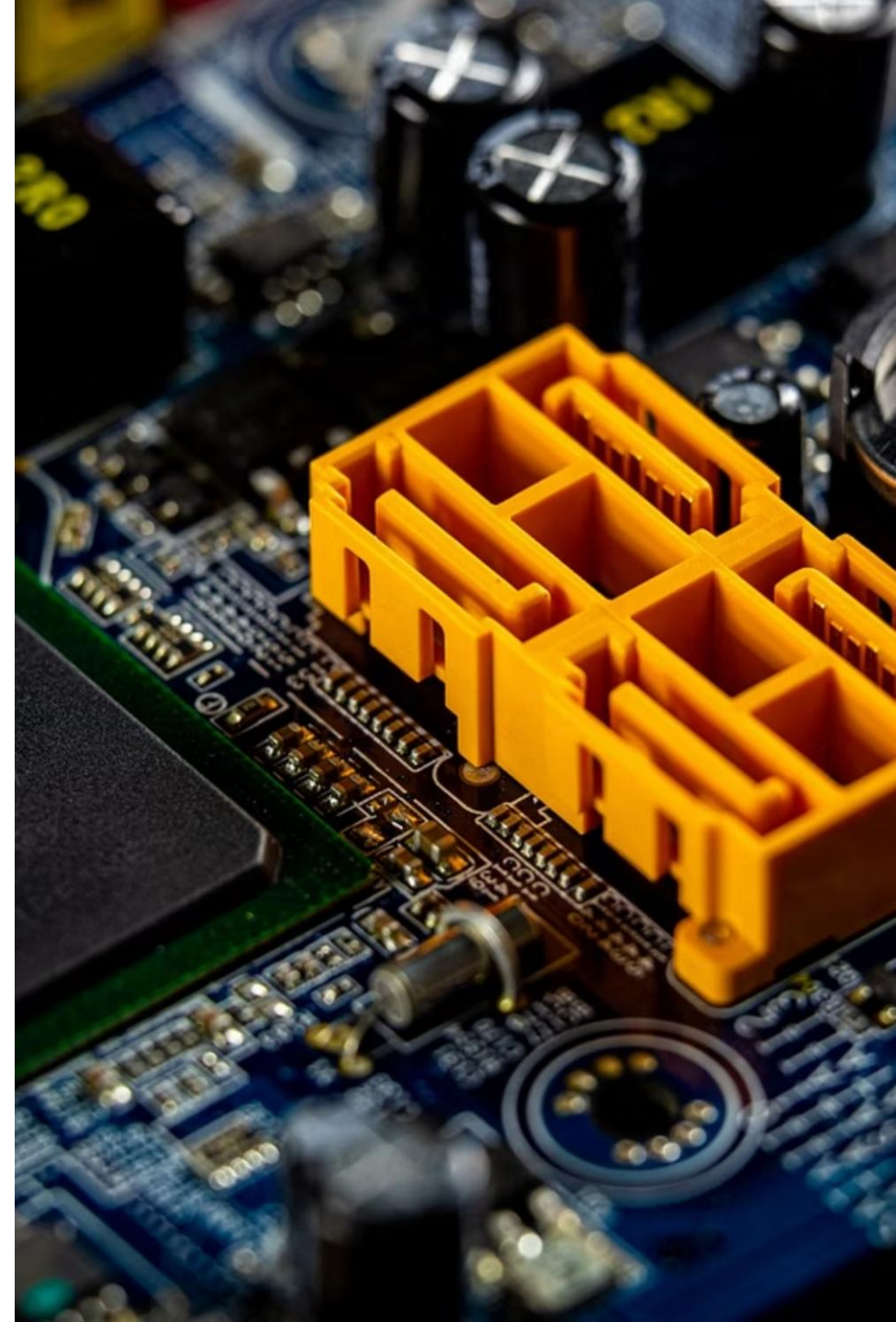


WCET Analysis Methods

Worst-case execution time in real time systems

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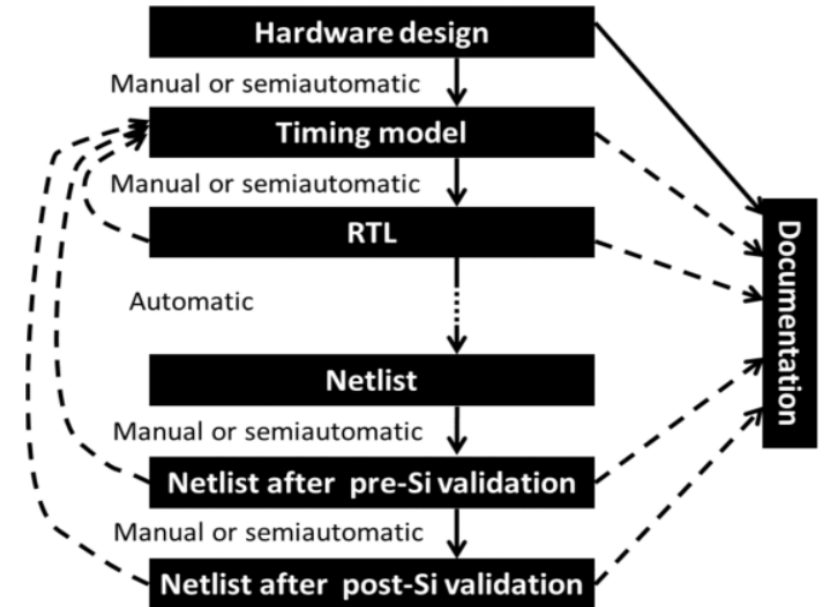
Why WCET matters

- Real-time systems must complete tasks before fixed deadlines
- WCET= Longest possible execution time
- A missed deadline = **timing failure**
- critical in: automotive, avionics, railway, space, industrial



Deterministic Methods

- **SDTA (Static Deterministic Timing Analysis):** Uses an abstract model of the hardware and a structural representation of the program. Calculates WCET analytically using program and hardware models without executing the program.
- **MBDTA (Measurement-Based Deterministic Timing Analysis):** Estimates WCET by running the program on real hardware and measuring execution times.
- **HYDTA (Hybrid Deterministic Timing Analysis):** Combines real execution measurements with static program analysis to improve WCET estimation accuracy.



Probabilistic methods

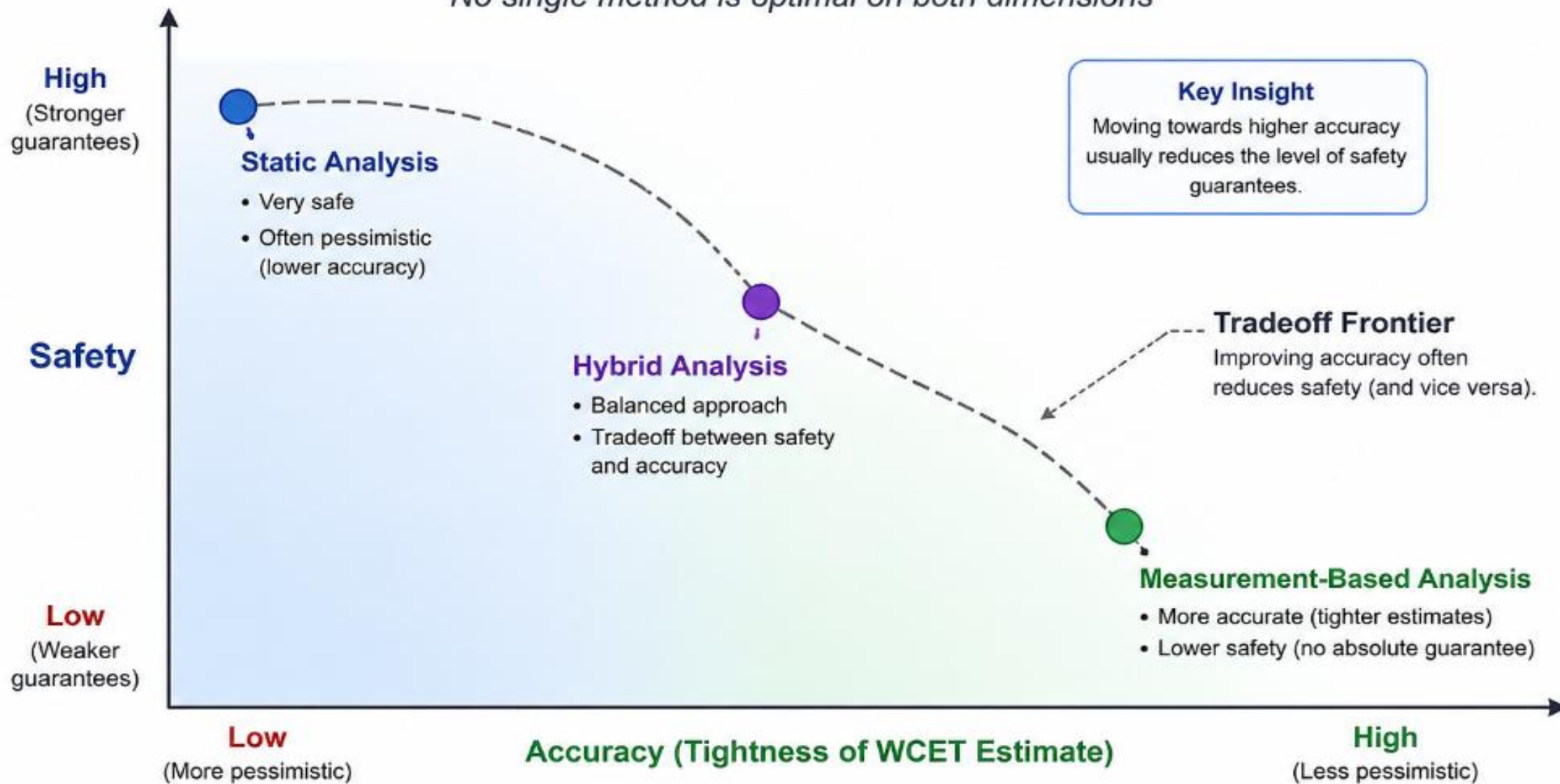
- **SPTA:** Predicts execution-time probabilities using mathematical models.
- **MBPTA:** Uses measurements to estimate worst-case timing probabilities.
- **HYPTA:** Combines measurements and analysis to obtain more reliable probabilistic WCET estimates.

Deterministic methods produce a single WCET value Eg. Unique/Scalar WCET (e.g., "A task will take at most X amount of time").

while probabilistic methods produce a probability of exceeding a given execution time. Probabilistic WCET (pWCET) distributions.

Tradeoff Between Safety and Accuracy

No single method is optimal on both dimensions



Limitations/ challenges

- **SDTA:**
 - inconsistencies in the timing model(TM) and the register transfer level(RTL).
 - Flow facts from the control flow graph defined at source code level but SDTA determined by low level object code.
- **MBDTA:**
 - Measurements need to be collected in a platform *identical* to those used at deployment.
 - Inputs used to collect measurements must - ideally - include the one leading to the WCET.
- **HYDTA:**
 - Complexity of the platform(multicore considerations)
- **PTA:**
 - Events have to be independent.

Flight Control Computer

- Aircraft control software must update control surfaces within a fixed deadline (e.g., 20 ms).
- SDTA analyzes all possible execution paths and processor behavior without running the program.
- The computed WCET is used during certification to prove that flight-control tasks always meet their deadlines.

Why not PTA?

Saying *"there is a 99.9999% chance the deadline will be met"* is often not sufficient for highly safety-critical flight-control functions.

Conclusion

- WCET analysis is essential for guaranteeing timing correctness in safety-critical real-time systems.
- Modern processor features such as caches, pipelines, and multicore resource sharing make accurate WCET estimation difficult. The main challenge is hardware unpredictability.
- Static, measurement-based, and hybrid methods each offer different tradeoffs between safety, accuracy and analysis effort.
- No single WCET analysis method is universally applicable to all systems and hardware platforms.

References

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